

Rotations, Non-Inertial Systems, and Rigid Body Motion

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Summary on Elementary Rotations

[source p. 1]

$$L_z = I \omega_z \quad (1)$$

$$\vec{N} = \vec{r} \times \vec{F} \quad (2)$$

$$T = \frac{I}{2} \omega_z^2 = \frac{\omega_z L_z}{2} \quad (3)$$

$$\frac{d\vec{L}}{dt} = \vec{N}^{(e)} \quad \left\{ \begin{array}{l} \text{INERTIAL FRAME} \\ \text{CM FRAME (ORIGIN AT CM, AXES PARALLEL TO INERTIAL AXES)} \end{array} \right. \quad (4)$$

Rolling without slipping condition

$$v_{\text{CM}x} + \omega_z R = 0 \quad (5)$$

$$(\omega_z < 0 \text{ for clockwise rotation}) \quad (6)$$

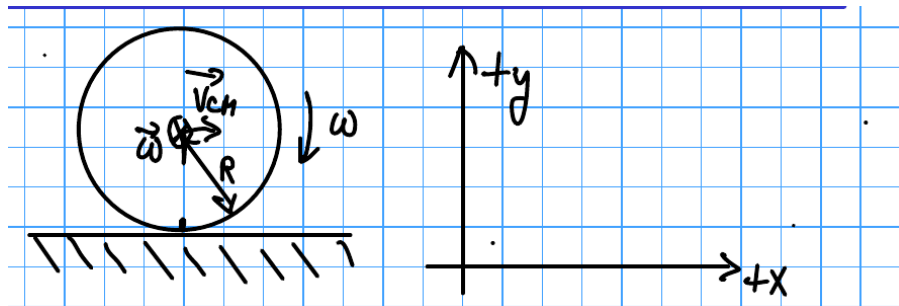


Figure 1: Rolling circle: $\vec{v}_{\text{CM}}$, $\vec{\omega}$ and the radius R to the contact point, with the sign convention of the $+x$, $+y$ axes drawn alongside.

Parallel Axis Theorem

$$I = \bar{I} + M a^2 \quad (7)$$

$$\bar{I} \equiv \text{Moment of inertia around CM} \quad (8)$$

$$a \equiv \text{Distance between axis of rotation and CM} \quad (9)$$

Separating CM Motion

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \sum_j \vec{r}'_j \times m_j \dot{\vec{r}}'_j \quad (10)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{1}{2} \sum_j m_j |\dot{\vec{r}}'_j|^2 \quad (11)$$

$$M = \sum_j m_j \quad (12)$$

$$\vec{R} \equiv \frac{\sum_j m_j \vec{r}_j}{M} \quad (13)$$

$$\vec{r}'_j \equiv \vec{r}_j - \vec{R} \quad \text{POSITION RELATIVE TO CM} \quad (14)$$

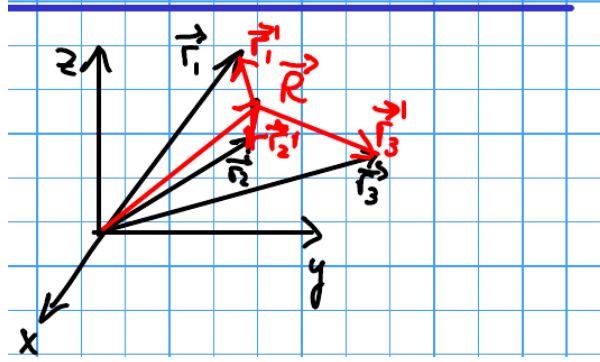


Figure 2: Laboratory position vectors $\vec{r}_1, \vec{r}_2, \vec{r}_3$ (black), the centre-of-mass vector $\vec{R}$ and the relative vectors $\vec{r}'_1, \vec{r}'_2, \vec{r}'_3$ (red).

For rigid body, considering rotation with angular velocity ω_z around CM:

$$L_z = (\vec{R} \times M \dot{\vec{R}})_z + I \omega_z \quad (16)$$

$$T = \frac{M}{2} |\dot{\vec{R}}|^2 + \frac{I}{2} \omega_z^2 \quad (17)$$

Proofs

CHANGING TO CM SYSTEM:

$$\begin{aligned} \vec{L} &= \sum_{i=1}^N \vec{r}_i \times m_i \dot{\vec{r}}_i = \sum_{i=1}^N (\vec{R} + \vec{r}'_i) \times m_i (\dot{\vec{R}} + \dot{\vec{r}}'_i) \\ &= \sum_i \vec{R} \times m_i \dot{\vec{R}} + \sum_i \vec{r}'_i \times m_i \dot{\vec{r}}'_i + \text{CROSS-TERMS} \end{aligned} \quad (18)$$

$$\boxed{\vec{r}_i = \vec{R} + \vec{r}'_i \quad \dot{\vec{r}}_i = \dot{\vec{R}} + \dot{\vec{r}}'_i} \quad (19)$$

CROSS-TERMS:

$$\begin{aligned}\sum_i \vec{r}'_i \times m_i \dot{\vec{R}} &= \sum_i m_i (\vec{r}_i - \vec{R}) \times \dot{\vec{R}} = \left[\underbrace{\sum_i m_i \vec{r}_i}_{M\vec{R}} - \underbrace{(\sum_i m_i) \vec{R}}_{M\vec{R}} \right] \times \dot{\vec{R}} \\ &= [M\vec{R} - M\vec{R}] \times \dot{\vec{R}} = \vec{0}\end{aligned}\quad (20)$$

$$\begin{aligned}\sum_i \vec{R} \times m_i \dot{\vec{r}}'_i &= \vec{R} \times \sum_i m_i (\dot{\vec{r}}_i - \dot{\vec{R}}) = \vec{R} \times \left[\underbrace{\sum_i m_i \dot{\vec{r}}_i}_{M\dot{\vec{R}}} - \underbrace{(\sum_i m_i) \dot{\vec{R}}}_{M\dot{\vec{R}}} \right] \\ &= \vec{R} \times [M\dot{\vec{R}} - M\dot{\vec{R}}] = \vec{0}\end{aligned}\quad (21)$$

THEN:

$$\boxed{\vec{L} = \vec{R} \times M\dot{\vec{R}} + \sum_i \vec{r}'_i \times m_i \dot{\vec{r}}'_i} \quad (22)$$

$$\Rightarrow \frac{d\vec{L}}{dt} = \vec{R} \times M\ddot{\vec{R}} + \underbrace{\sum_i \vec{r}'_i \times m_i \ddot{\vec{r}}'_i}_{d\vec{L}'/dt} \quad \text{TIME DERIVATIVE} \quad (23)$$

BUT ALSO:

$$\frac{d\vec{L}}{dt} = \vec{N}^{(e)} = \sum_i \vec{r}_i \times \vec{F}_i^{(e)} \quad (24)$$

$$M\ddot{\vec{R}} = \sum_i \vec{F}_i^{(e)} \quad \text{Newton's 2nd law for particle system} \quad (25)$$

$$\frac{d\vec{L}}{dt} = \vec{R} \times \sum_i \vec{F}_i^{(e)} + \sum_i \vec{r}'_i \times \vec{F}_i^{(e)} = \vec{R} \times M\ddot{\vec{R}} + \sum_{i=1}^N \vec{r}'_i \times \vec{F}_i^{(e)} \quad (26)$$

$$\boxed{\frac{d\vec{L}'}{dt} = \sum_{i=1}^N \vec{r}'_i \times \vec{F}_i^{(e)}} \quad (27)$$

Eq (27) is the special case of Eq (4) in the CM Frame.¹

Circular motion in a plane; rotating coordinate systems

[source p. 2]

$$\begin{aligned}\vec{v} &= \vec{\omega} \times \vec{r} \\ \vec{v} \perp \vec{r} \quad \vec{r} \cdot \dot{\vec{r}} &= 0\end{aligned}$$

$$\hat{e}_i \cdot \hat{e}_j = \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

¹*Editorial note.* The numbers (1)–(27) above belong to this summary sheet alone; they are the lecturer's own and are reproduced verbatim. He wrote no (15). A second, unrelated sequence beginning again at (1) starts on source p. 9 below.

Rotating coordinate system

$$0 = \frac{d\delta_{ij}}{dt} = \frac{d\hat{e}_i \cdot \hat{e}_j}{dt} \implies \begin{cases} 0 = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt}, & i = j \\ 0 = \hat{e}_i \cdot \frac{d\hat{e}_j}{dt} + \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}, & i \neq j \end{cases}$$

$(\hat{e}_j)_{j=1,2,3}$ IS A BASIS $\implies$ WE CAN WRITE $d\hat{e}_i/dt$ AS A LINEAR COMBINATION

$$\boxed{\frac{d\hat{e}_i}{dt} = \sum_{j=1}^3 \Omega_{ij} \hat{e}_j \quad \Omega_{ij} = \hat{e}_j \cdot \frac{d\hat{e}_i}{dt}}$$

$$\left(\text{USE THAT } \hat{e}_j \cdot \sum_{i=1}^3 v_i \hat{e}_i = \sum_{i=1}^3 v_i \hat{e}_j \cdot \hat{e}_i = v_j \right)$$

$$\langle \varphi_i | \left(\sum_j c_j |\varphi_j\rangle \right) = \sum_j c_j \langle \varphi_i | \varphi_j \rangle = c_i$$

For $i \neq j$:

$$0 = \Omega_{ij} + \Omega_{ji} \implies \Omega_{ji} = -\Omega_{ij}$$

$$\Omega_{ii} = \hat{e}_i \cdot \frac{d\hat{e}_i}{dt} = 0$$

$$\begin{pmatrix} 0 & \Omega_{12} & -\Omega_{31} \\ -\Omega_{12} & 0 & \Omega_{23} \\ \Omega_{31} & -\Omega_{23} & 0 \end{pmatrix} \quad \omega_1 \equiv \Omega_{23} \quad \omega_2 = \Omega_{31} \quad \omega_3 = \Omega_{12}$$

$$\frac{d\hat{e}_1}{dt} = \Omega_{12}\hat{e}_2 + \Omega_{13}\hat{e}_3 = \Omega_{12}\hat{e}_2 - \Omega_{31}\hat{e}_3 = \omega_3\hat{e}_2 - \omega_2\hat{e}_3 = \vec{\omega} \times \hat{e}_1 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_1 = (\omega_1\hat{e}_1 + \omega_2\hat{e}_2 + \omega_3\hat{e}_3) \times \hat{e}_1 = -\omega_2\hat{e}_3 + \omega_3\hat{e}_2$$

$$\frac{d\hat{e}_2}{dt} = \Omega_{21}\hat{e}_1 + \Omega_{23}\hat{e}_3 = -\Omega_{12}\hat{e}_1 + \Omega_{23}\hat{e}_3 = -\omega_3\hat{e}_1 + \omega_1\hat{e}_3 = \vec{\omega} \times \hat{e}_2 \quad \checkmark$$

$$\vec{\omega} \times \hat{e}_2 = (\omega_1\hat{e}_1 + \omega_2\hat{e}_2 + \omega_3\hat{e}_3) \times \hat{e}_2 = \omega_1\hat{e}_3 - \omega_3\hat{e}_1$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3, \quad \hat{e}_3 \times \hat{e}_2 = -\hat{e}_1$$

$$\frac{d\hat{e}_3}{dt} = \vec{\omega} \times \hat{e}_3$$

Unit vectors and the Levi-Civita symbol

$$\hat{x} \times \hat{y} = \hat{z} \quad \hat{y} \times \hat{z} = \hat{x} \quad \hat{z} \times \hat{x} = \hat{y}$$

$$\hat{e}_1 \equiv \hat{x} \quad \hat{e}_2 \equiv \hat{y} \quad \hat{e}_3 \equiv \hat{z}$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 \quad (\text{AND CYCLICALLY}) \quad \hat{e}_2 \times \hat{e}_3 = \hat{e}_1 \quad \hat{e}_3 \times \hat{e}_1 = \hat{e}_2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & ijk \text{ CYC } 123 \\ -1 & ijk \text{ CYC } 213 \\ 0 & \text{if any repeated index} \end{cases}$$

$$\begin{aligned}
\epsilon_{123} &= \epsilon_{231} = \epsilon_{312} = +1 \\
\epsilon_{213} &= \epsilon_{132} = \epsilon_{321} = -1 \\
\epsilon_{jji} &= \epsilon_{ijj} = \epsilon_{jij} = 0 \\
\hat{e}_i \times \hat{e}_j &= \sum_{\ell} \epsilon_{ij\ell} \hat{e}_{\ell}
\end{aligned}$$

Rotating reference frames

[source p. 3]

Fixed inertial reference frame $\hat{e}_1^0 \hat{e}_2^0 \hat{e}_3^0$; rotating (non-inertial) reference frame $\hat{e}_1 \hat{e}_2 \hat{e}_3$.

$$\begin{aligned}
\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} &= \frac{d}{dt} \left(\sum_{i=1}^3 b_i^0 \hat{e}_i^0 \right) = \sum_{i=1}^3 \dot{b}_i^0 \hat{e}_i^0 \\
\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} &= \frac{d}{dt} \left(\sum_{i=1}^3 b_i \hat{e}_i \right) = \sum_{i=1}^3 \dot{b}_i \hat{e}_i + \sum_{i=1}^3 b_i \dot{\hat{e}}_i \quad \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} \equiv \sum_{i=1}^3 \frac{db_i}{dt} \hat{e}_i \\
\dot{\hat{e}}_i &= \frac{d\hat{e}_i}{dt} = \vec{\omega} \times \hat{e}_i
\end{aligned}$$

For rotations,

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \sum_{i=1}^3 b_i (\vec{\omega} \times \hat{e}_i)$$

$$\left. \frac{d\vec{b}}{dt} \right|_{\text{inertial}} = \left. \frac{d\vec{b}}{dt} \right|_{\text{body}} + \vec{\omega} \times \vec{b}$$

Rigid body rotations

$\vec{r}_P$: position of point P in the body frame (CONSTANT IN BODY FRAME). Rigid body with one fixed point.

$$T = \sum_P \frac{1}{2} m_P v_P^2 = \sum_P \frac{1}{2} m_P (\vec{\omega} \times \vec{r}_P)^2$$

Levi-Civita symbol

$$\epsilon_{ijk} = \begin{cases} +1 & ijk \text{ CYC } 123 \\ -1 & ijk \text{ CYC } 213 \\ 0 & \text{if any repeated index} \end{cases}$$

$$\begin{aligned}
\epsilon_{123} &= \epsilon_{231} = \epsilon_{312} = +1 \\
\epsilon_{213} &= \epsilon_{132} = \epsilon_{321} = -1 \\
\epsilon_{jji} &= \epsilon_{ijj} = \epsilon_{jij} = 0
\end{aligned}$$

$$\hat{e}_1 \times \hat{e}_2 = \hat{e}_3 = \sum_{\ell} \epsilon_{12\ell} \hat{e}_{\ell} \quad \checkmark \quad \sum_{\ell} \epsilon_{12\ell} \hat{e}_{\ell} = \hat{e}_3$$

$$\hat{e}_i \times \hat{e}_j = \sum_{\ell} \epsilon_{ij\ell} \hat{e}_{\ell} \quad \left(= \epsilon_{ij\ell} \hat{e}_{\ell} \text{ NOTATION} \right)$$

$$\vec{a} \times \vec{b} = \left(\sum_i a_i \hat{e}_i \right) \times \left(\sum_j b_j \hat{e}_j \right) = \sum_{ij\ell} a_i b_j \epsilon_{ij\ell} \hat{e}_{\ell} = \sum_{ijk} \epsilon_{ijk} a_i b_j \hat{e}_k \quad (\vec{a} \times \vec{b})_k = \sum_{ij} \epsilon_{ijk} a_i b_j$$

Also

$$\sum_i \epsilon_{jki} \epsilon_{\ell mi} = \sum_i \epsilon_{ijk} \epsilon_{ilm} = \delta_{j\ell} \delta_{km} - \delta_{jm} \delta_{\ell k}$$

$$\epsilon_{ijk} = \epsilon_{jki} = \epsilon_{kij} \quad (\text{CYCLIC})$$

$$\begin{aligned} (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) &= \left(\sum_{ijk} A_i B_j \epsilon_{ijk} \hat{e}_k \right) \cdot \left(\sum_{lmn} C_{\ell} D_m \epsilon_{lmn} \hat{e}_n \right) = \sum_{ijk\ell m} \epsilon_{ijk} \epsilon_{\ell mk} A_i B_j C_{\ell} D_m \\ &= \sum_{ijk\ell m} (\delta_{i\ell} \delta_{jm} - \delta_{im} \delta_{j\ell}) A_i B_j C_{\ell} D_m = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \end{aligned}$$

In summary

[source p. 4]

$$T = \sum_P \frac{1}{2} m_P v_P^2 = \sum_P \frac{1}{2} m_P (\vec{\omega} \times \vec{r}_P)^2 = \sum_P \frac{m_P}{2} [\omega^2 r_P^2 - (\vec{\omega} \cdot \vec{r}_P)^2]$$

$$\begin{aligned} (\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) &= (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}) \\ \implies (\vec{\omega} \times \vec{r}_P) \cdot (\vec{\omega} \times \vec{r}_P) &= (\vec{\omega} \cdot \vec{\omega})(\vec{r}_P \cdot \vec{r}_P) - (\vec{\omega} \cdot \vec{r}_P)(\vec{\omega} \cdot \vec{r}_P) \end{aligned}$$

$$\begin{aligned} T &= \frac{1}{2} \sum_P m_P \left[r_P^2 \sum_{i=1}^3 \omega_i \omega_i - \left(\sum_{i=1}^3 \omega_i r_{Pi} \right) \left(\sum_{j=1}^3 \omega_j r_{Pj} \right) \right] \\ &= \frac{1}{2} \sum_P m_P \sum_{ij=1}^3 \left[r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right] \omega_i \omega_j \end{aligned}$$

$$T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j$$

$$I_{ij} \equiv \sum_P m_P \left[r_P^2 \delta_{ij} - r_{Pi} r_{Pj} \right]$$

Moment of inertia
tensor (SYMMETRIC)

$$\vec{L} = \sum_P m_P \vec{r}_P \times \vec{v}_P = \sum_P m_P \vec{r}_P \times (\vec{\omega} \times \vec{r}_P)$$

Aside. Using summation notation:

$$\begin{aligned} [\vec{A} \times (\vec{B} \times \vec{C})]_i &= \epsilon_{ijk} A_j \epsilon_{klm} B_l C_m \stackrel{\epsilon_{ijk} = \epsilon_{kij}}{=} \epsilon_{kij} \epsilon_{klm} A_j B_l C_m = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) A_j B_l C_m \\ &= (\vec{A} \cdot \vec{C}) B_i - (\vec{A} \cdot \vec{B}) C_i \end{aligned}$$

$$[\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)]_i = (\vec{r}_P \cdot \vec{r}_P) \omega_i - (\vec{r}_P \cdot \vec{\omega}) r_{Pi}$$

$$L_i = \sum_P m_P (\vec{r}_P \times \vec{v}_P)_i = \sum_P m_P [\vec{r}_P \times (\vec{\omega} \times \vec{r}_P)]_i = \sum_P m_P \left(r_P^2 \omega_i - \sum_{j=1}^3 r_{Pi} r_{Pj} \omega_j \right)$$

$$L_i = \sum_{j=1}^3 \underbrace{\left(\sum_P m_P [r_P^2 \delta_{ij} - r_{Pi} r_{Pj}] \right)}_{= I_{ij}} \omega_j = \sum_{j=1}^3 I_{ij} \omega_j$$

$$L_i = \sum_j I_{ij} \omega_j$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L}$$

$$T = \frac{1}{2} \sum_{ij=1}^3 I_{ij} \omega_i \omega_j$$

Principal axes

[source p. 5]

$$I_{ij} = I_{ji} = I_{ji}^* \implies \text{HERMITIAN} \implies \text{CAN BE DIAGONALIZED} \quad \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}$$

Directions that diagonalize I_{ij} are called PRINCIPAL AXES.

Examples

Homogeneous CUBE OR SPHERE:

$$\begin{pmatrix} I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{pmatrix} = I \mathbf{1}_{3 \times 3}$$

$$I = \sum_P m_P (\text{distance to axis})_P^2 = \sum_P m_P r_{P\perp}^2$$

$$I_3 \sim \# Ma^2 \quad I_1 \sim \# Mb^2$$

$$a \ll b \implies I_3 \ll I_1 \quad a \gg b \implies I_3 \gg I_1$$

Caveat 1. The numerical coefficient in front of Ma^2 and Mb^2 is struck out in the source and is not legible. Only the scaling is asserted; no value has been supplied here.

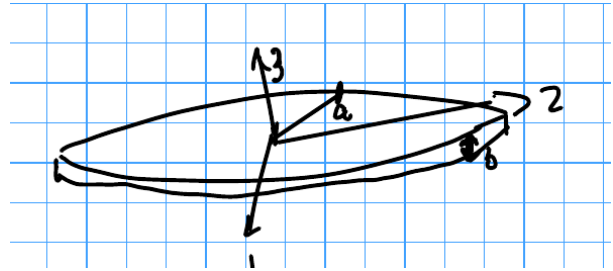
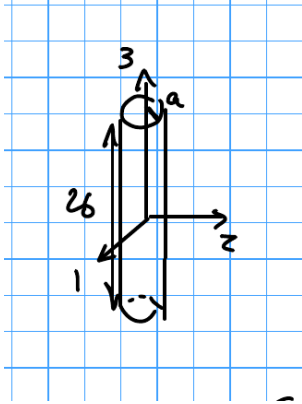


Figure 3: Left: long thin rod of radius a and length $2b$ about the body axes 1, 2, 3; $a \ll b \Rightarrow I_1 = I_2 > I_3$. Right: flat disc, $a \gg b \Rightarrow I_1 = I_2 < I_3$.

$$\begin{aligned}
 I_{ij} &= \sum_P m_P [\delta_{ij} r_P^2 - r_{Pi} r_{Pj}] \\
 I_{33} &= \sum_P m_P [\underbrace{\delta_{33}}_1 r_P^2 - r_{P3} r_{P3}] \\
 r_{P1} r_{P1} + r_{P2} r_{P2} + r_{P3} r_{P3} - r_{P3} r_{P3} &= r_{P\perp}^2 \\
 x x + y y + z z - z z &= x^2 + y^2 = \rho^2
 \end{aligned}$$

More irregular: I_1, I_2, I_3 all different.

$$\begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \longleftrightarrow I_{ij} = I_i \delta_{ij} \quad L_i = \sum_{j=1}^3 I_{ij} \omega_j = \sum_{j=1}^3 I_i \delta_{ij} \omega_j = I_i \omega_i$$

$$\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} = \begin{pmatrix} I_1 \omega_1 \\ I_2 \omega_2 \\ I_3 \omega_3 \end{pmatrix}$$

$$T = \frac{1}{2} \vec{\omega} \cdot \vec{L} = \frac{1}{2} \sum_{i=1}^3 \omega_i L_i = \frac{1}{2} \sum_{i=1}^3 \omega_i I_i \omega_i = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

$$L_i = I_i \omega_i$$

$$T = \frac{1}{2} \sum_{i=1}^3 I_i \omega_i^2$$

Parallel Axis Theorem

[source p. 6]

($\bar{I}_{ij}$ M of I w.r.t. CM)

$$\begin{array}{c} I_{ij} \\ \uparrow \\ \text{ORIGIN O} \end{array} = \begin{array}{c} \bar{I}_{ij} \\ \uparrow \\ \text{ORIGIN CM} \end{array} + M [\delta_{ij} a^2 - a_i a_j]$$

Elementary version: $I = \bar{I} + Md^2$.

Rotation w.r.t. an arbitrarily moving point

No longer look at a point in the body fixed in space.

- INERTIAL FRAME
- CM FRAME $\begin{cases} \text{MAY NOT BE INERTIAL IF } \ddot{\vec{R}} \neq \vec{0} \\ \text{AXES} \parallel \text{AXES OF INERTIAL FRAME} \end{cases}$
- BODY FRAME $\begin{cases} \text{ORIGIN AT } \vec{R} \Rightarrow \text{NONINERTIAL IF } \ddot{\vec{R}} \neq \vec{0} \\ \text{AXES } 1\,2\,3 \text{ ROTATING W.R.T. } 1_0\,2_0\,3_0 \end{cases}$

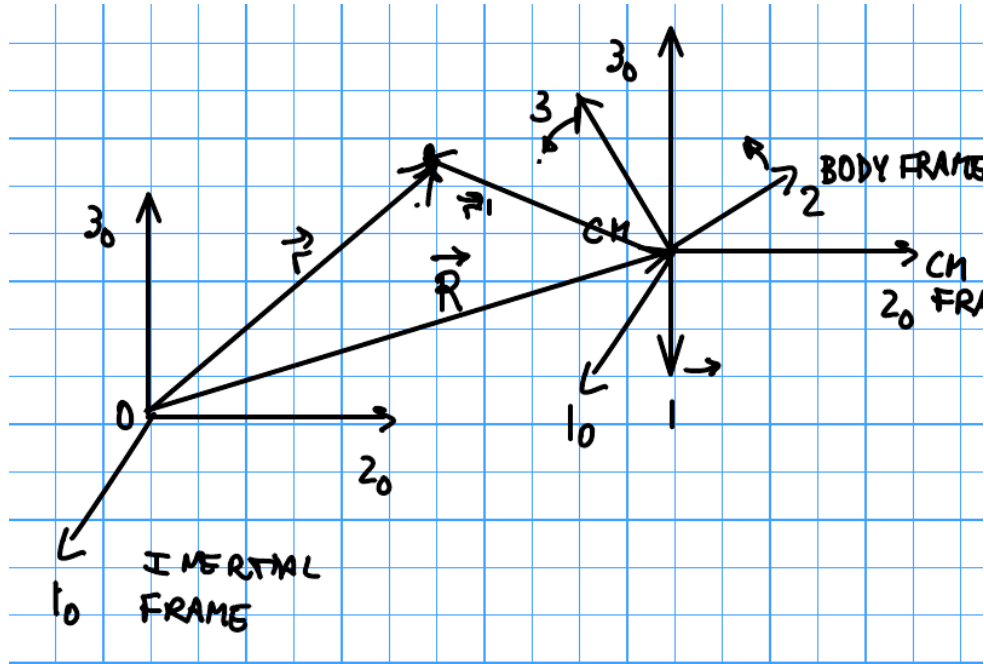


Figure 4: The three frames: inertial ($1_0\,2_0\,3_0$ at O), the CM frame (axes parallel to the inertial ones), and the body frame ($1\,2\,3$, rotating with the body), with $\vec{r} = \vec{R} + \vec{r}'$.

$$\vec{r} = \vec{R} + \vec{r}'$$

$$\left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{CM}}$$

$$\left. \frac{d\vec{r}'}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \left. \frac{d\vec{r}'}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{r}'$$

$$\text{IF RIGID BODY} \quad \left. \frac{d\vec{r}}{dt} \right|_{\text{INERTIAL}} = \frac{d\vec{R}}{dt} + \vec{\omega} \times \vec{r}'$$

For any system of particles:

$$M = \sum_P m_P \quad \vec{R} \equiv \begin{array}{c} \text{CENTER} \\ \text{OF MASS} \\ \text{POSITION} \end{array} \equiv \frac{\sum_P m_P \vec{r}_P}{\sum_P m_P}$$

$$T = \frac{1}{2} M \dot{\vec{R}}^2 + T' \quad T' = \sum_P \frac{m_P}{2} \dot{\vec{r}}_P'^2$$

$$\vec{L} = \vec{R} \times M \dot{\vec{R}} + \vec{L}' \quad \vec{L}' = \sum_P \vec{r}_P' \times m_P \dot{\vec{r}}_P'$$

(the primed quantities are MEASURED IN FRAME WITH ORIGIN AT CM — the CM FRAME).

For rigid body $\dot{\vec{r}}_P' = \vec{\omega} \times \vec{r}_P'$:

$$\begin{aligned} \vec{L} &= \vec{R} \times M \dot{\vec{R}} + \underbrace{\sum_P \vec{r}_P' \times m_P (\vec{\omega} \times \vec{r}_P')}_{\text{CALCULATED BEFORE}} & L_i &= \sum_j \bar{I}_{ij} \omega_j \\ T &= \frac{1}{2} M |\dot{\vec{R}}|^2 + \underbrace{\sum_P \frac{m_P}{2} (\vec{\omega} \times \vec{r}_P')^2}_{\text{CALCULATED BEFORE}} & T' &= \frac{1}{2} \sum_{ij} \bar{I}_{ij} \omega_i \omega_j \end{aligned}$$

Euler's Equations

[source p. 7]

$$\frac{d\vec{L}}{dt} = \vec{N} \quad \text{IF} \quad \left\{ \begin{array}{l} \times \text{ INERTIAL FRAME} \\ \times \text{ CM FRAME (ORIGIN AT CM MAY BE} \\ \text{ACCELERATED BUT IT STAYS PARALLEL} \\ \text{TO INERTIAL FRAME)} \end{array} \right\} \Leftrightarrow \begin{array}{l} \text{"TRANSLATED} \\ \text{BUT NOT ROTATED"} \end{array}$$

$$\vec{N} = \frac{d\vec{L}}{dt} = \left. \frac{d\vec{L}}{dt} \right|_{\text{BODY}} + \vec{\omega} \times \vec{L}$$

$$N_1 = \left. \frac{dL_1}{dt} \right|_{\text{BODY}} + (\vec{\omega} \times \vec{L})_1 = \left. \frac{dL_1}{dt} \right|_{\text{BODY}} + \omega_2 L_3 - \omega_3 L_2$$

with $L_i = I_i \omega_i$ and $\dot{I}_i \Big|_{\text{BODY}} = 0$ (rigid body $\Rightarrow \dot{\vec{r}}_P = \vec{0}$ in body frame $\Rightarrow \dot{I}_i \Big|_{\text{BODY}} = 0$):

$$N_1 = I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} + \omega_2 I_3 \omega_3 - \omega_3 I_2 \omega_2 = I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} + \omega_2 \omega_3 (I_3 - I_2)$$

$$\boxed{\begin{array}{l} I_1 \left. \frac{d\omega_1}{dt} \right|_{\text{BODY}} = \omega_2 \omega_3 (I_2 - I_3) + N_1 \\ I_2 \left. \frac{d\omega_2}{dt} \right|_{\text{BODY}} = \omega_3 \omega_1 (I_3 - I_1) + N_2 \\ I_3 \left. \frac{d\omega_3}{dt} \right|_{\text{BODY}} = \omega_1 \omega_2 (I_1 - I_2) + N_3 \end{array}}$$

EULER'S
EQUATIONS

Applications: Kater's Pendulum

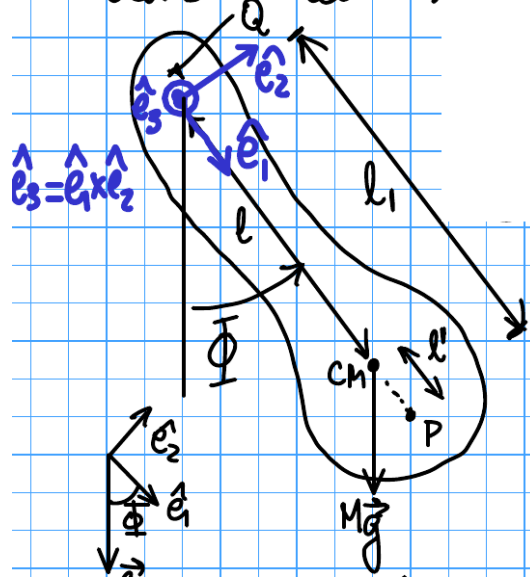


Figure 5: Kater's pendulum. Suspension point Q ; body triad $\hat{e}_1, \hat{e}_2, \hat{e}_3 = \hat{e}_1 \times \hat{e}_2$ (blue); l from Q to the CM, l_1 to the percussion centre P at l' beyond the CM; generalized coordinate $\bar{\Phi}$.

$n = 1$ degree of freedom $\iff$ angle $\bar{\Phi} \equiv$ gen coord. NO FRICTION. $\omega_3 = \dot{\bar{\Phi}}$.

In Lab frame:

$$I \dot{\omega}_3 = \frac{dL_3}{dt} = N_3 = -Mgl \sin \bar{\Phi}$$

$$\vec{N}_g = l \hat{e}_1 \times M\vec{g} = l \hat{e}_1 \times Mg(\cos \bar{\Phi} \hat{e}_1 - \sin \bar{\Phi} \hat{e}_2) = -Mgl \sin \bar{\Phi} \hat{e}_3$$

$$\vec{g} = g(\cos \bar{\Phi} \hat{e}_1 - \sin \bar{\Phi} \hat{e}_2)$$

SMALL ANGLE $\sin \bar{\Phi} \approx \bar{\Phi}$:

$$I \ddot{\bar{\Phi}} \approx -Mgl \bar{\Phi} \quad \ddot{\bar{\Phi}} \approx -\frac{Mgl}{I} \bar{\Phi} \quad \ddot{\bar{\Phi}} = -\Omega^2 \bar{\Phi} \quad \Omega = \left(\frac{Mgl}{I}\right)^{1/2} \text{ FREQ. OF OSCILLATION}$$

Parallel axis theorem

[source p. 8]

$$I = \bar{I} + Ml^2 = M\bar{k}^2 + Ml^2 = M(\bar{k}^2 + l^2) = Mk^2$$

DEFINE RADIUS OF GYRATION k BY $I \equiv Mk^2$; DEFINE $l_1 \equiv \frac{k^2}{l} = l + \frac{\bar{k}^2}{l}$.

$$\Omega_Q^2 = \frac{Mgl}{M(\bar{k}^2 + l^2)} = \frac{g}{l_1} \quad [\Omega^2] = \frac{L/T^2}{L} = \frac{1}{T^2} \implies [\Omega] = \frac{1}{T}$$

COMPARE WITH SIMPLE PENDULUM $\Omega = \sqrt{g/l}$ (point mass):

$$\bar{I} = 0 \implies \bar{k} = 0 \quad l_1 = l + \frac{\bar{k}^2}{l} = l$$

$$\bar{I} = \int d^3r \rho(\vec{r}) (r_1^2 + r_2^2) \quad \begin{matrix} \uparrow \uparrow \\ \text{distances measured w.r.t. CM} \end{matrix}$$

(for a point mass $r_1 = r_2 = 0$)

$$\bar{k}^2 = \frac{\bar{I}}{M} = \frac{\int d^3r \rho(\vec{r}) (r_1^2 + r_2^2)}{\int d^3r \rho(\vec{r})} \iff \begin{matrix} \text{Average squared distance to axis of rotation} \\ \text{weighted by the mass density } \rho(\vec{r}) \end{matrix}$$

Suspend the pendulum from point P ("Percussion center") at distance l_1 from Q_1 along the direction that goes through CM:

$$\Omega_Q^2 = \frac{Mgl}{M(\bar{k}^2 + l^2)} = \frac{g}{l_1} \quad \Omega_P^2 = \frac{gl'}{\bar{k}^2 + l'^2} = \frac{g(l_1 - l)}{\bar{k}^2 + (l_1 - l)^2} = \frac{g\bar{k}^2/l}{\bar{k}^2 + \bar{k}^4/l^2} = \frac{gl}{l^2 + \bar{k}^2} = \Omega_Q^2$$

$$l_1 = l + \frac{\bar{k}^2}{l} \quad l_1 - l = \frac{\bar{k}^2}{l} \quad (l_1 - l)^2 = \frac{\bar{k}^4}{l^2}$$

Torque-Free Motion in 3D: Symmetric Top

$$I_1 = I_2 \neq I_3, \quad \Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1}.$$

$$\begin{aligned} I_1 \frac{d\omega_1}{dt} \Big|_{\text{BODY}} &= \omega_2 \omega_3 (I_2 - I_3) \implies \dot{\omega}_1 = - \left[\omega_3 \frac{I_3 - I_2}{I_1} \right] \omega_2 \\ I_2 \frac{d\omega_2}{dt} \Big|_{\text{BODY}} &= \omega_3 \omega_1 (I_3 - I_1) \implies \dot{\omega}_2 = \left[\omega_3 \frac{I_3 - I_1}{I_2} \right] \omega_1 \\ I_3 \frac{d\omega_3}{dt} \Big|_{\text{BODY}} &= \omega_1 \omega_2 (I_1 - I_2) \implies \end{aligned}$$

$$\begin{aligned} \dot{\omega}_1 &= -\Omega \omega_2 \\ \dot{\omega}_2 &= \Omega \omega_1 \\ \dot{\omega}_3 &= 0 \end{aligned}$$

[source p. 9]

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} \iff \vec{\omega} \text{ rotates with angular velocity } \Omega \hat{e}_3 \implies \text{PRECESSION}$$

$$\dot{\vec{\omega}} = \Omega \hat{e}_3 \times \vec{\omega} = \Omega \hat{e}_3 \times (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2 + \omega_3 \hat{e}_3) = \Omega (\omega_1 \hat{e}_2 - \omega_2 \hat{e}_1)$$

$$\dot{\omega}_1 = -\Omega \omega_2 \quad \dot{\omega}_2 = \Omega \omega_1 \quad \dot{\omega}_3 = 0$$

$\vec{\omega}$ rotates around $\hat{e}_3$ with ang. speed Ω . If initially $\vec{\omega} = \omega(\sin \lambda \hat{e}_1 + \cos \lambda \hat{e}_3)$, then $\vec{\omega} \in$ plane of $\hat{e}_1$ and $\hat{e}_3$:

$$\omega_1 = \omega \sin \lambda \cos(\Omega t) \quad \omega_2 = \omega \sin \lambda \sin(\Omega t) \quad \omega_3 = \omega \cos \lambda$$

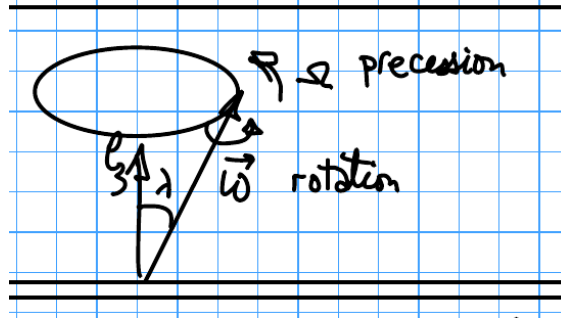


Figure 6: $\vec{\omega}$ (rotation) sweeps a cone of half-angle λ about the body axis $\hat{e}_3$ at the precession rate Ω .

Euler Angles

$\hat{e}_1^0 \hat{e}_2^0 \hat{e}_3^0$ INERTIAL SET OF UNIT VECTORS (NOT ROTATING); $\hat{e}_1 \hat{e}_2 \hat{e}_3$ BODY SET OF UNIT VECTORS (ROTATING WITH BODY). The three-frame construction of Figure 4 is redrawn here.

Following Ch5 F&W (Skipping a lot)

We need 6 coordinates to describe positions of all points in a RIGID BODY:

- CM POSITION $\vec{R} \iff X Y Z$;
- 3 coord to fix orientation of $\hat{e}_1 \hat{e}_2 \hat{e}_3$ w.r.t. $\hat{e}_1^0 \hat{e}_2^0 \hat{e}_3^0$.

WE CHOOSE EULER ANGLES: α, β, γ .

- $\hat{e}_\beta \perp \hat{e}_3^0, \hat{e}_3$
- ORIENTATION OF NEW NORTH POLE $\longrightarrow \beta$ POLAR, α AZIMUTHAL
- γ SPECIFIES ADDITIONAL ROTATION AROUND $\hat{e}_3$
- α, β, γ CAN BE CHOSEN INDEPENDENTLY AND THEY COMPLETELY SPECIFY ORIENTATION OF BODY FRAME $\implies$ **USE $\alpha \beta \gamma$ AS GENERALIZED COORD**

$$T = \frac{1}{2} (I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2) \quad (1)$$

$$\vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\beta} \hat{e}_\beta + \dot{\gamma} \hat{e}_3 \quad (2)$$

$$\hat{e}_\beta = \cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1 \quad (3)$$

$$\hat{e}_3^0 = \cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2] \quad (4)$$

$$\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$$

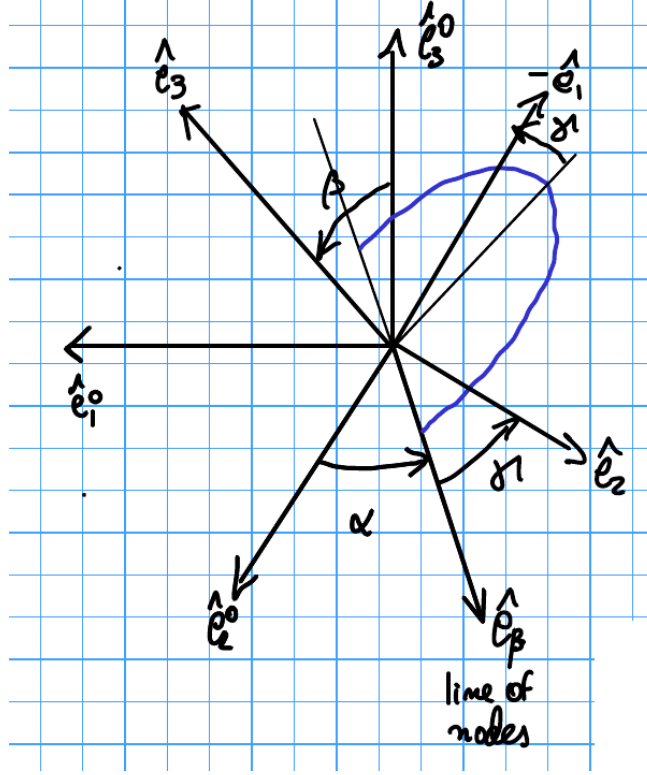


Figure 7: Definition of the Euler angles. Space triad $\hat{e}_1^0, \hat{e}_2^0, \hat{e}_3^0$; body triad $\hat{e}_1, \hat{e}_2, \hat{e}_3$ (note $-\hat{e}_1$ as drawn); α azimuthal, β polar, γ the additional rotation about $\hat{e}_3$; $\hat{e}_\beta$ is the line of nodes.

[source p. 10]

NOTE: Infinitesimal rot.

$$R_\alpha = \mathbf{1} + M_\alpha \dot{\alpha} dt \quad (6)$$

$$R_\beta = \mathbf{1} + M_\beta \dot{\beta} dt \quad (7)$$

$$R_\alpha R_\beta = \mathbf{1} + (M_\alpha \dot{\alpha} + M_\beta \dot{\beta}) dt + \mathcal{O}((dt)^2) \quad (8)$$

$$R_\beta R_\alpha = \mathbf{1} + (M_\beta \dot{\beta} + M_\alpha \dot{\alpha}) dt + \mathcal{O}((dt)^2) \quad (9)$$

EQUAL TO ORDER dt — the $\mathcal{O}((dt)^2)$ terms are struck out

$$\Rightarrow \text{INFINITESIMAL ROTATIONS ADD,} \quad \vec{\omega} = \vec{\omega}_\alpha + \vec{\omega}_\beta + \vec{\omega}_\gamma \quad (10)$$

$$(10) \Rightarrow (2)$$

Kinetic energy — General case

$$\begin{aligned} T &= \frac{I_1}{2} \omega_1^2 + \frac{I_2}{2} \omega_2^2 + \frac{I_3}{2} \omega_3^2 \\ &= \frac{I_1}{2} (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma)^2 + \frac{I_2}{2} (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma)^2 + \frac{I_3}{2} (\dot{\gamma} + \dot{\alpha} \cos \beta)^2 \end{aligned} \quad (11)$$

$$\text{Symmetric Top:} \quad I_1 = I_2 \quad (12)$$

$$\boxed{T = \frac{I_1}{2}(\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2}(\dot{\gamma} + \dot{\alpha} \cos \beta)^2} \quad (13)$$

Torque-Free Motion of symmetric top

$$\boxed{L = T = \frac{I_1}{2}(\dot{\beta}^2 + \dot{\alpha}^2 \sin^2 \beta) + \frac{I_3}{2}(\dot{\gamma} + \dot{\alpha} \cos \beta)^2} \quad (14)$$

α, β, γ Gen Coord $\implies$ 3 Lagrange Eqns.

$$2 \text{ CYCLIC COORDINATES: } \alpha, \gamma \quad (15)$$

$\implies$ 2 CONST OF MOTION:

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16)$$

$$p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17)$$

WE WILL SHOW LATER THAT:

$$p_\alpha = \vec{L} \cdot \hat{e}_3^0 \quad (18)$$

$$p_\gamma = \vec{L} \cdot \hat{e}_3 \quad (19)$$

We compute now

$$p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17)$$

and notice that, using (5),

$$\boxed{p_\gamma = I_3 \omega_3 = \vec{L} \cdot \hat{e}_3} \quad \checkmark \quad \text{AS ANTICIPATED} \quad (19)$$

Also

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (16)$$

$$\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \quad (20)$$

Compare with

$$\vec{L} \cdot \hat{e}_3^0 = \overbrace{\left(I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \right)}^{(20)} \cdot \overbrace{\left(\cos \beta \hat{e}_3 + \sin \beta [\cos \gamma (-\hat{e}_1) + \sin \gamma \hat{e}_2] \right)}^{(4)} \quad (21)$$

$$= I_1 \left[(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) (-\sin \beta \cos \gamma) + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) (\sin \beta \sin \gamma) \right] + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (22)$$

$$= I_1 \dot{\alpha} \sin^2 \beta (\cos^2 \gamma + \sin^2 \gamma) + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta \quad (23)$$

$$\boxed{\vec{L} \cdot \hat{e}_3^0 = I_1 \dot{\alpha} \sin^2 \beta + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta = p_\alpha} \quad \checkmark \quad (18)$$

$$p_\beta = \frac{\partial L}{\partial \dot{\beta}} = I_1 \dot{\beta} \quad (24)$$

NOT A CONSTANT (IN PRINCIPLE) BECAUSE β APPEARS EXPLICITLY IN THE LAGRANGIAN.

$$\frac{\partial L}{\partial \beta} = I_1 \sin \beta \cos \beta \dot{\alpha}^2 - I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \dot{\alpha} \sin \beta = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \quad (25)$$

Then

$$\boxed{\dot{p}_\beta = \frac{\partial L}{\partial \beta} = \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3)} \quad (\text{However, see later}) \quad (26)$$

Compare p_β with

$$\vec{L} \cdot \hat{e}_\beta = \overbrace{\left(I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \right)}^{(20)} \cdot \overbrace{(\cos \gamma \hat{e}_2 + \sin \gamma \hat{e}_1)}^{(3)}$$

Recall

$$\vec{\omega} = (-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \hat{e}_1 + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \hat{e}_2 + (\dot{\gamma} + \dot{\alpha} \cos \beta) \hat{e}_3 \quad (5)$$

$$\vec{L} \cdot \hat{e}_\beta = I_1 \left[(-\dot{\alpha} \sin \beta \cos \gamma + \dot{\beta} \sin \gamma) \sin \gamma + (\dot{\alpha} \sin \beta \sin \gamma + \dot{\beta} \cos \gamma) \cos \gamma \right] \quad (27)$$

$$= I_1 \dot{\beta} (\sin^2 \gamma + \cos^2 \gamma) = I_1 \dot{\beta} \quad (28)$$

$$\implies \boxed{\vec{L} \cdot \hat{e}_\beta = p_\beta} \quad \checkmark \quad (29)$$

Description in Inertial (or CM) Frame

$$\text{No torque} \quad (30)$$

$$\dot{\vec{L}} = \vec{0} \quad (31)$$

$$\iff \begin{cases} \text{CHOOSE } \hat{e}_3^0 \parallel \vec{L} & (32) \\ |\vec{L}| \text{ doesn't change} & (33) \end{cases}$$

$$\implies \vec{L} \cdot \hat{e}_3^0 = |\vec{L}| \quad (34)$$

$$\text{AND} \quad \vec{L} = |\vec{L}| \hat{e}_3^0 \quad (35)$$

We know that $\vec{L} \cdot \hat{e}_3 = p_\gamma$ is a constant of the motion:

$$\underbrace{\vec{L} \cdot \hat{e}_3}_{= p_\gamma} = \underbrace{|\vec{L}|}_{= \text{CONST}} \hat{e}_3^0 \cdot \hat{e}_3 \quad (36)$$

$$\implies \hat{e}_3^0 \cdot \hat{e}_3 \text{ CONSTANT} \quad (37)$$

$$\text{CONST} = \hat{e}_3^0 \cdot \hat{e}_3 = \cos(\text{angle between } \hat{e}_3 \text{ and } \hat{e}_3^0) = \cos \beta \quad (38)$$

$$\beta \text{ IS A CONSTANT OF THE MOTION} \quad (39)$$

$$\Rightarrow \dot{\beta} = 0 \quad (40)$$

$$\Rightarrow \boxed{p_\beta = I_1 \dot{\beta} = \vec{L} \cdot \hat{e}_\beta = 0} \quad (41)$$

[margin: For this particular choice of axes, p_β and $\vec{L} \cdot \hat{e}_\beta$ are zero!!]

$$\left. \begin{array}{l} \vec{L} \cdot \hat{e}_\beta = 0 \\ \vec{L} \neq 0, \quad |\hat{e}_\beta| = 1 \neq 0 \end{array} \right\} \quad (41) \Rightarrow \vec{L} \perp \hat{e}_\beta \quad (42)$$

$$\Rightarrow \boxed{\vec{L} \text{ is in the plane of } \hat{e}_3 \text{ and } \hat{e}_3^0} \quad (43)$$

CONSTANTS: $I_1, I_3, \beta, I_3\omega_3 = L_3, p_\alpha$.

$$p_\alpha = \frac{\partial L}{\partial \dot{\alpha}} = \underbrace{I_1 \sin^2 \beta \dot{\alpha} + I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \cos \beta}_{(16)} \stackrel{(5)}{=} \underline{I_1 \sin^2 \beta \dot{\alpha} + I_3 \omega_3 \cos \beta} \quad (44)$$

$$\Rightarrow \boxed{\dot{\alpha} \text{ CONSTANT}} \quad (45)$$

$$\left. \begin{array}{l} p_\gamma = \frac{\partial L}{\partial \dot{\gamma}} = I_3 (\dot{\gamma} + \dot{\alpha} \cos \beta) \quad (17) \\ I_3, p_\gamma, \dot{\alpha}, \beta \text{ CONSTANT} \end{array} \right\} \Rightarrow \boxed{\dot{\gamma} \text{ CONSTANT}} \quad (46)$$

$$\alpha = \dot{\alpha}_0 t \quad (47)$$

$$\gamma = \dot{\gamma}_0 t \quad (48)$$

$\dot{\alpha}_0, \dot{\gamma}_0$ CONSTANTS

[source p. 12]

$$0 \stackrel{(41)}{=} \dot{p}_\beta = \frac{\partial L}{\partial \beta} \stackrel{(26)}{=} \dot{\alpha} \sin \beta (I_1 \dot{\alpha} \cos \beta - I_3 \omega_3) \Rightarrow \boxed{\dot{\alpha} \cos \beta = \frac{I_3 \omega_3}{I_1}} \quad (49)$$

$$(5) \Rightarrow \omega_3 = \dot{\gamma} + \dot{\alpha} \cos \beta \stackrel{(49)}{=} \dot{\gamma} + \frac{I_3}{I_1} \omega_3 \Rightarrow \boxed{\dot{\gamma} = \omega_3 \frac{I_1 - I_3}{I_1} = -\Omega} \quad (51)$$

$$\Omega \equiv \omega_3 \frac{I_3 - I_1}{I_1} \quad (50)$$

defined when looking at body frame evolution of $\vec{\omega}$ a couple of lectures before

$$(2) + (40) \Rightarrow \boxed{\vec{\omega} = \dot{\alpha} \hat{e}_3^0 + \dot{\gamma} \hat{e}_3} \quad (52)$$

$$\Rightarrow \boxed{\hat{e}_3, \hat{e}_3^0, \vec{\omega}, \vec{L} \text{ are in one plane}} \quad (53)$$

Consider cases where $I_1 = I_2 < I_3$ or $I_1 = I_2 > I_3$:

$$\vec{L} = I_1 (\omega_1 \hat{e}_1 + \omega_2 \hat{e}_2) + I_3 \omega_3 \hat{e}_3 \quad (20)$$

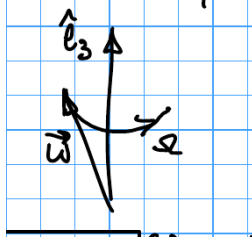


Figure 8: $\vec{\omega}$ and the precession rate Ω relative to the body axis $\hat{e}_3$.

SIMPLE PENDULUM:

$$\ddot{\Theta} = -\frac{g}{l} \sin \Theta$$

THUMBTACK:

$$(\text{angle})'' = - \begin{bmatrix} \text{AWFUL} \\ \text{MESS} \end{bmatrix} \sin(\text{angle})$$
